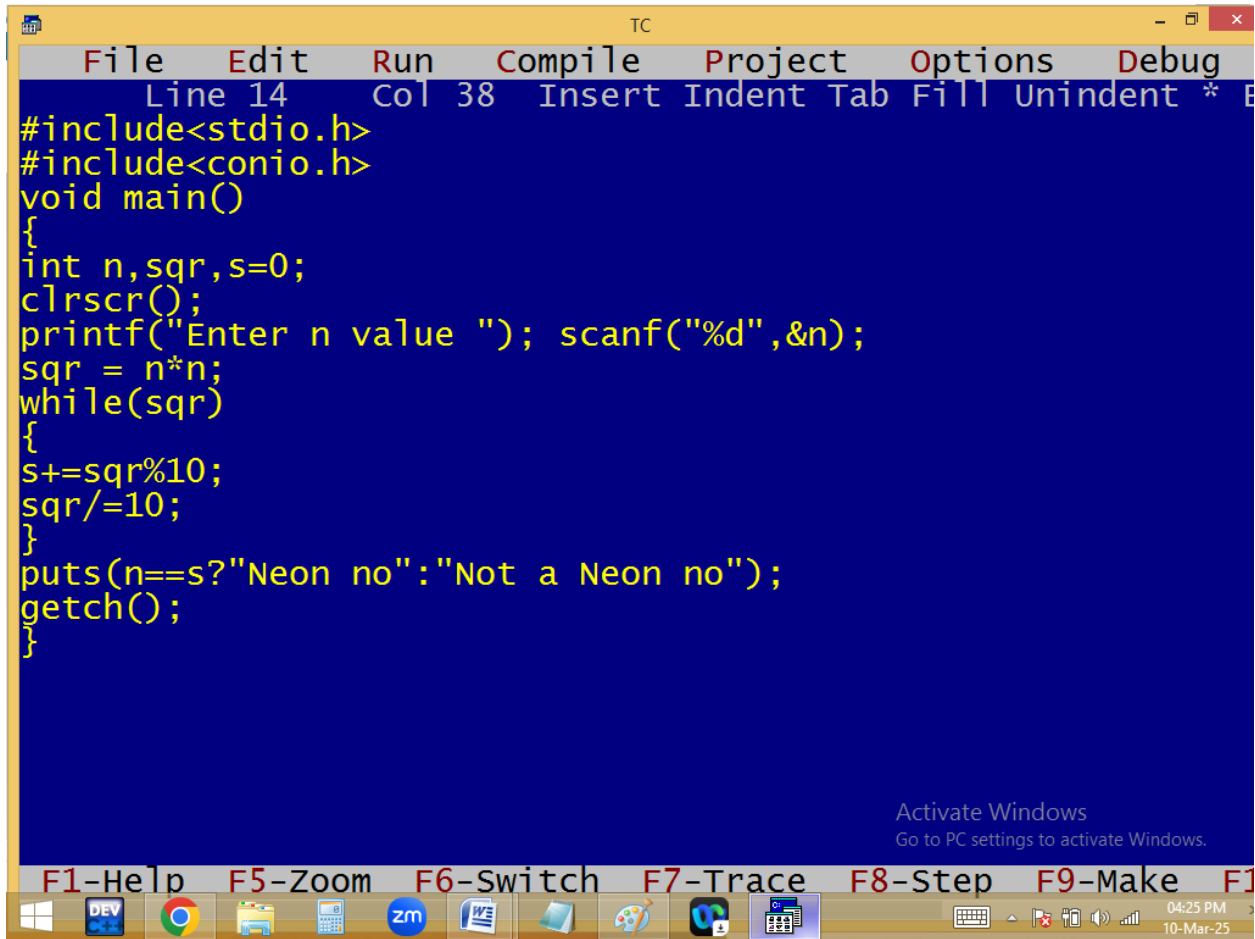


Neon number – sqr of given no digits sum is equal to given no

Eg: 9 → $9*9=81$ → $8+1=9$



The screenshot shows a Turbo C++ (TC) IDE window. The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates 'Line 14 Col 38' and provides shortcuts for Insert, Indent, Tab, Fill, Unindent, and a wildcard *. The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int n,sqr,s=0;
    clrscr();
    printf("Enter n value "); scanf("%d",&n);
    sqr = n*n;
    while(sqr)
    {
        s+=sqr%10;
        sqr/=10;
    }
    puts(n==s?"Neon no":"Not a Neon no");
    getch();
}
```

At the bottom of the IDE, there is a toolbar with function key shortcuts: F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, and F10-Run. The Windows taskbar is visible at the very bottom, showing the Start button, taskbar icons for DEV, Chrome, File Explorer, Calculator, Zoom, Word, and other applications, along with the system clock showing 04:25 PM on 10-Mar-25.



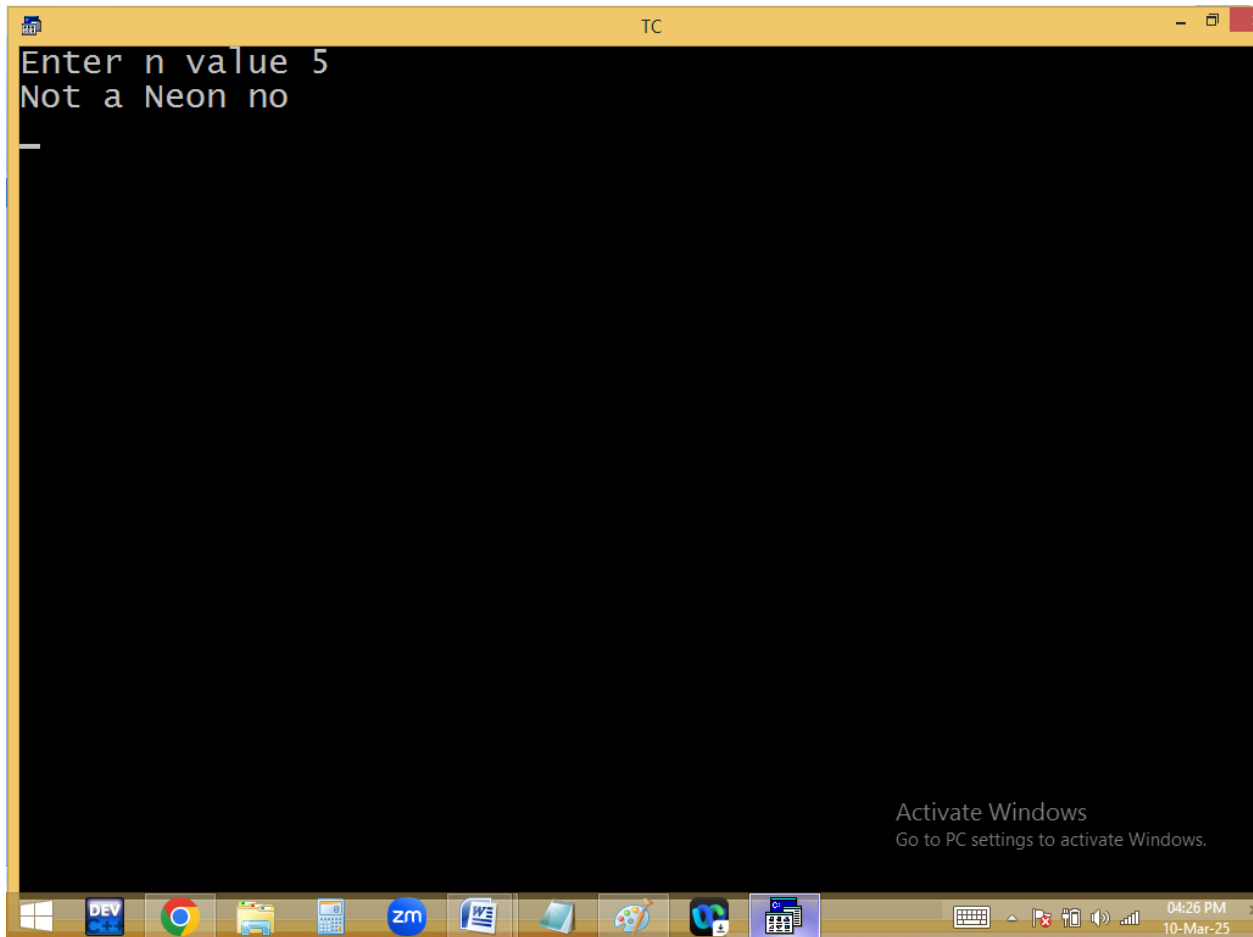
TC

Enter n value 9
Neon no

Activate Windows
Go to PC settings to activate Windows.



04:25 PM
10-Mar-25



```
sqr = n*n;
```

```
    81
```

```
while(sqr)
```

```
{
```

```
    s=s+sqr%10;
```

```
    sqr/=10;
```

```
}
```

```
puts( n==s ?neon: not);
```

$$\frac{n}{9}$$

$$\frac{m}{9}$$

$$\frac{sqr}{n \times n = 81}$$

$$\frac{8}{0+1}$$

8

$$\frac{8}{9}$$

harshad number / niven number

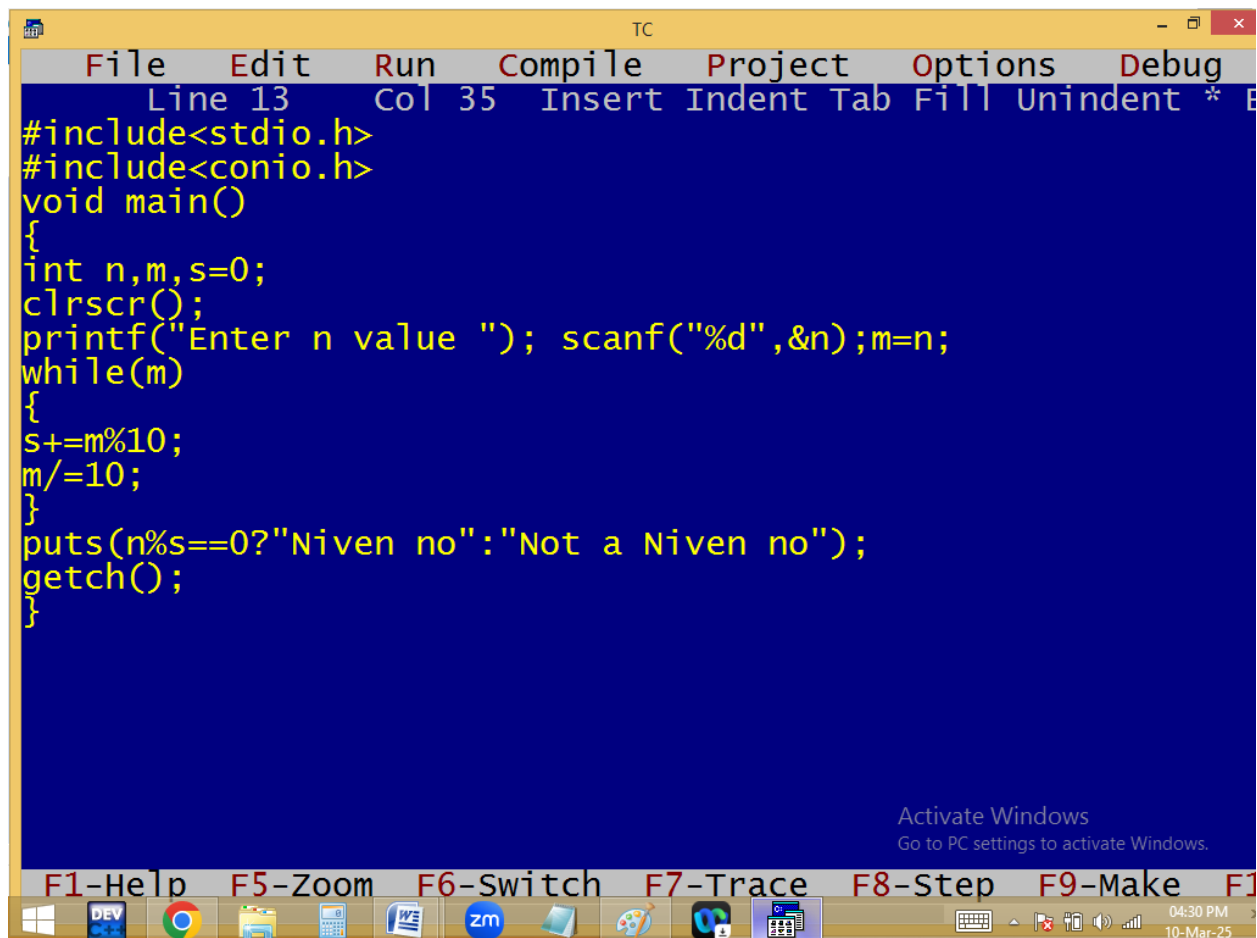
**18 is a harshad number because $1 + 8 = 9$,
and 18 is divisible by 9**

1729 is a harshad number because $1 + 7 + 2 + 9 = 19$, and 1729 is divisible by 19

2025 is a harshad number because $2 + 0 + 2 + 5 = 9$, and 2025 is divisible by 9

Other names

$1234=1+2+3+4=10 \leftarrow$ not a harshad no



```
TC
File Edit Run Compile Project Options Debug
Line 13 Col 35 Insert Indent Tab Fill Unindent * B
#include<stdio.h>
#include<conio.h>
void main()
{
int n,m,s=0;
clrscr();
printf("Enter n value "); scanf("%d",&n);m=n;
while(m)
{
s+=m%10;
m/=10;
}
puts(n%s==0?"Niven no":"Not a Niven no");
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run

04:30 PM 10-Mar-25



TC

Enter n value 18
Niven no

Activate Windows
Go to PC settings to activate Windows.



04:30 PM
10-Mar-25



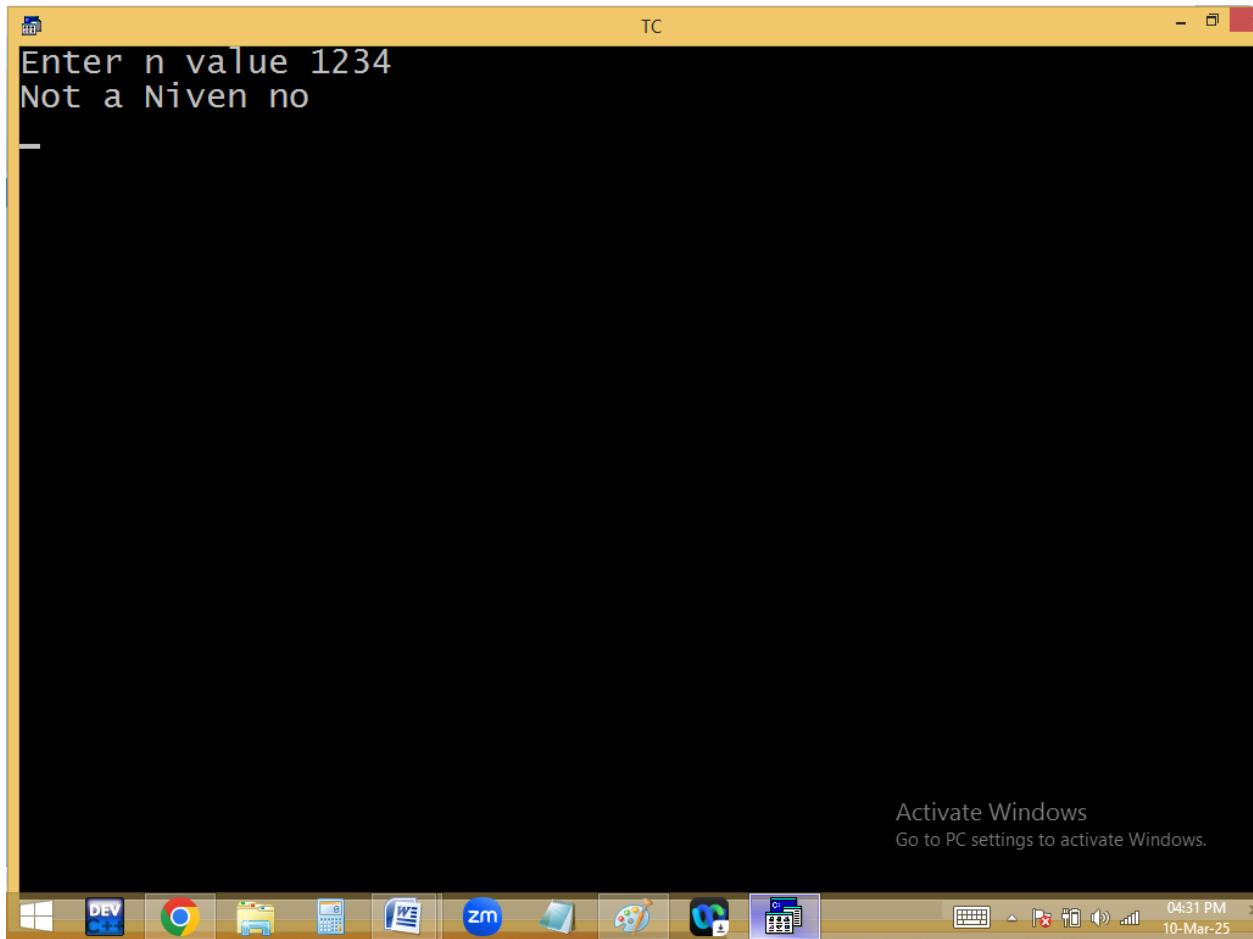
TC

Enter n value 1729
Niven no

Activate Windows
Go to PC settings to activate Windows.



04:31 PM
10-Mar-25



$$\begin{array}{r}
 n \\
 \hline
 18
 \end{array}
 \quad
 \begin{array}{r}
 m \\
 \hline
 18
 \end{array}
 \begin{array}{l}
 \% 10 = 8 + 0 = 8 \\
 + \% 10 = 1 + 8 = 9 \\
 \square
 \end{array}$$

```

while(m)
{
s+=m%10; m/=10;
}

```

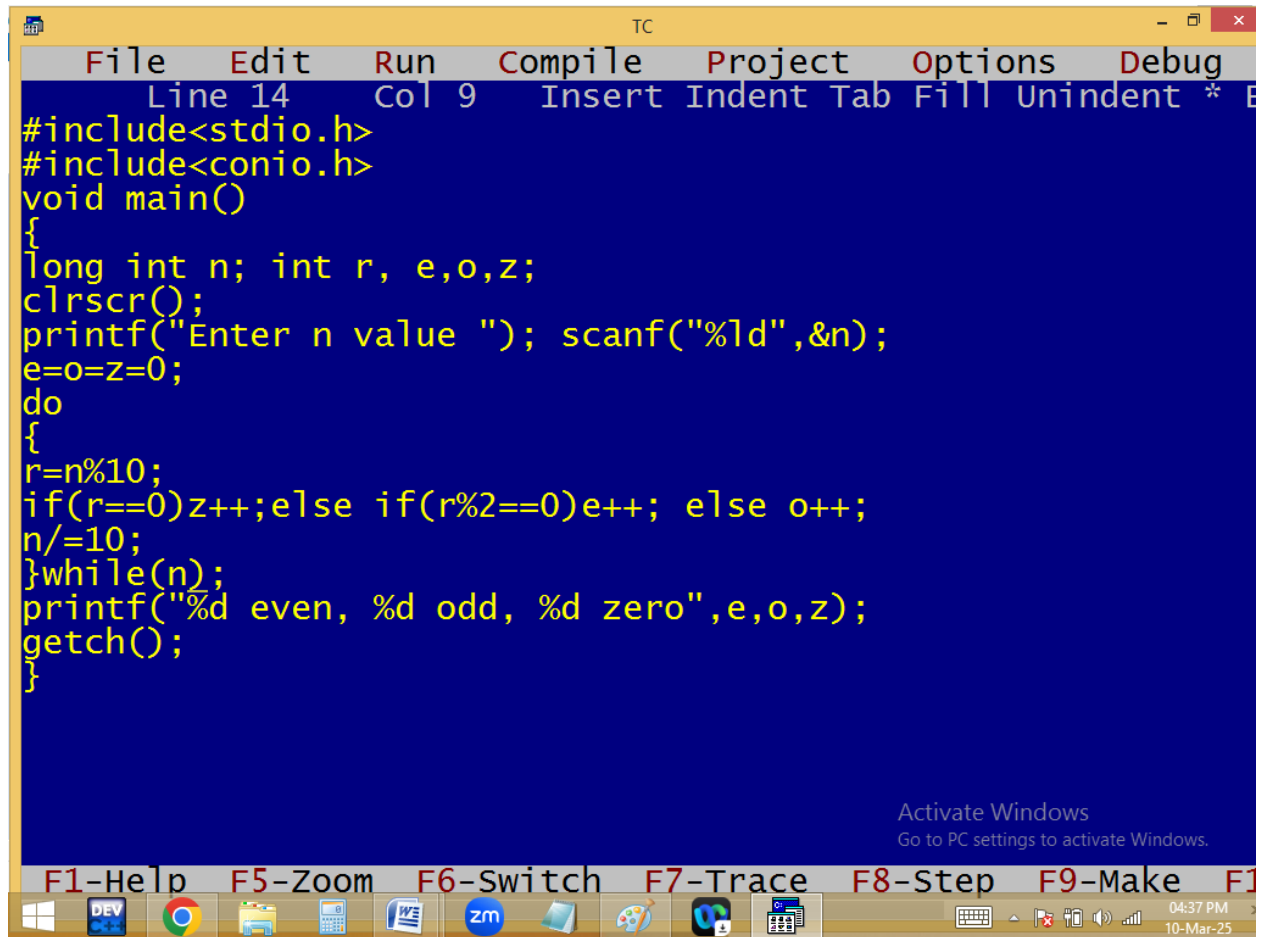
```

if( n%s==0) p(niven);
else not

```

Finding no of even/odd/0 digits in given no

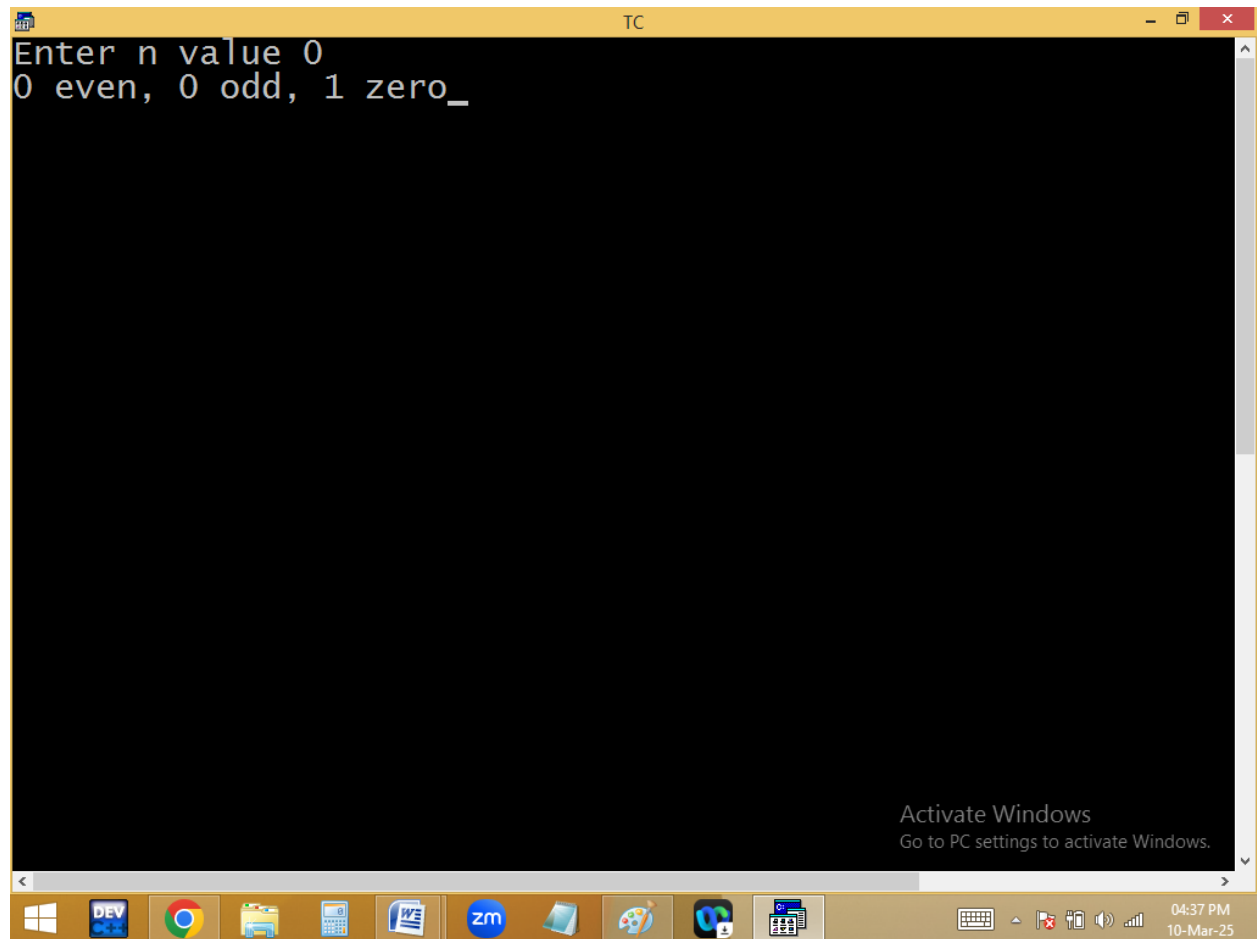
eg: 1002 → 1 even, 1 odd, 2 zero

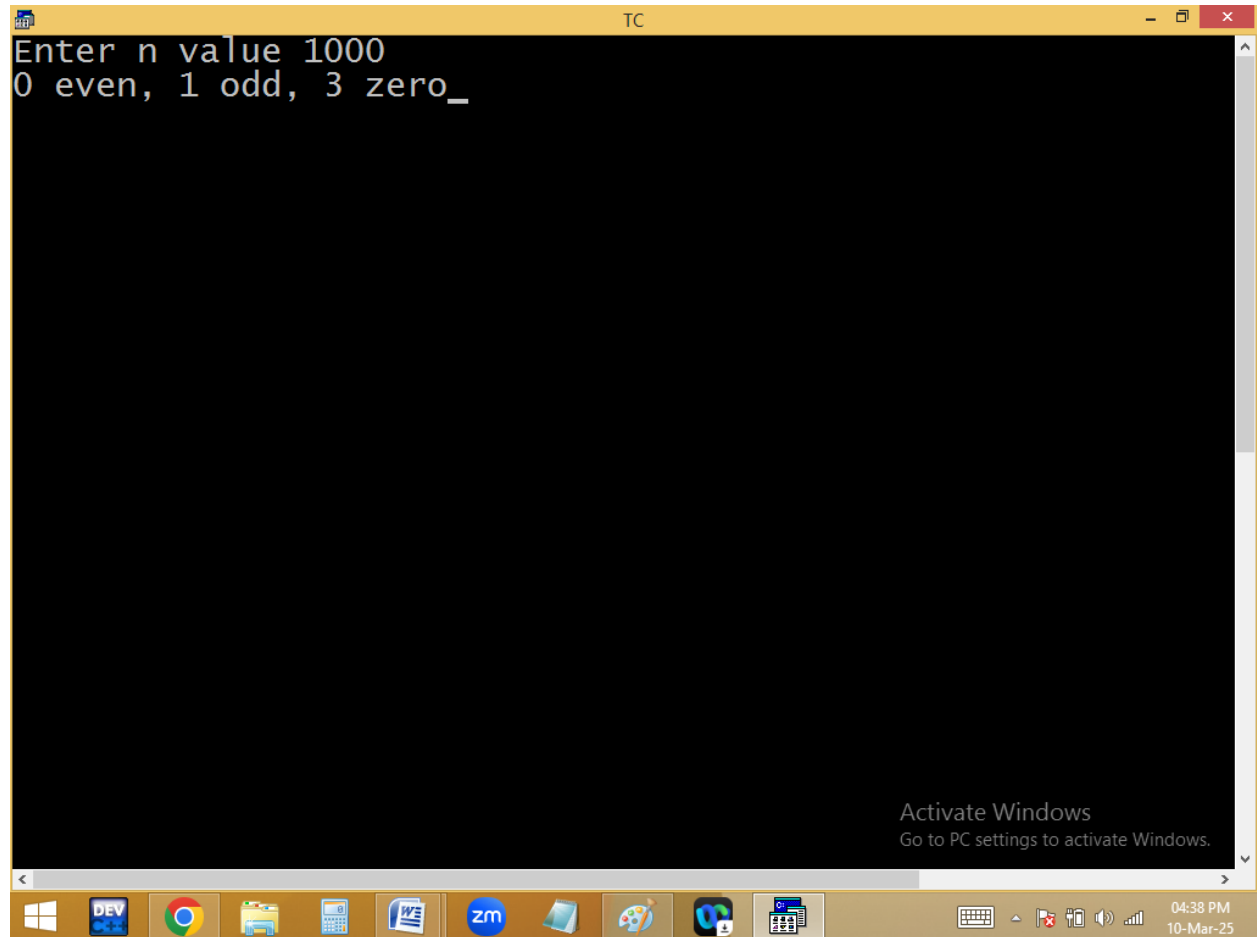


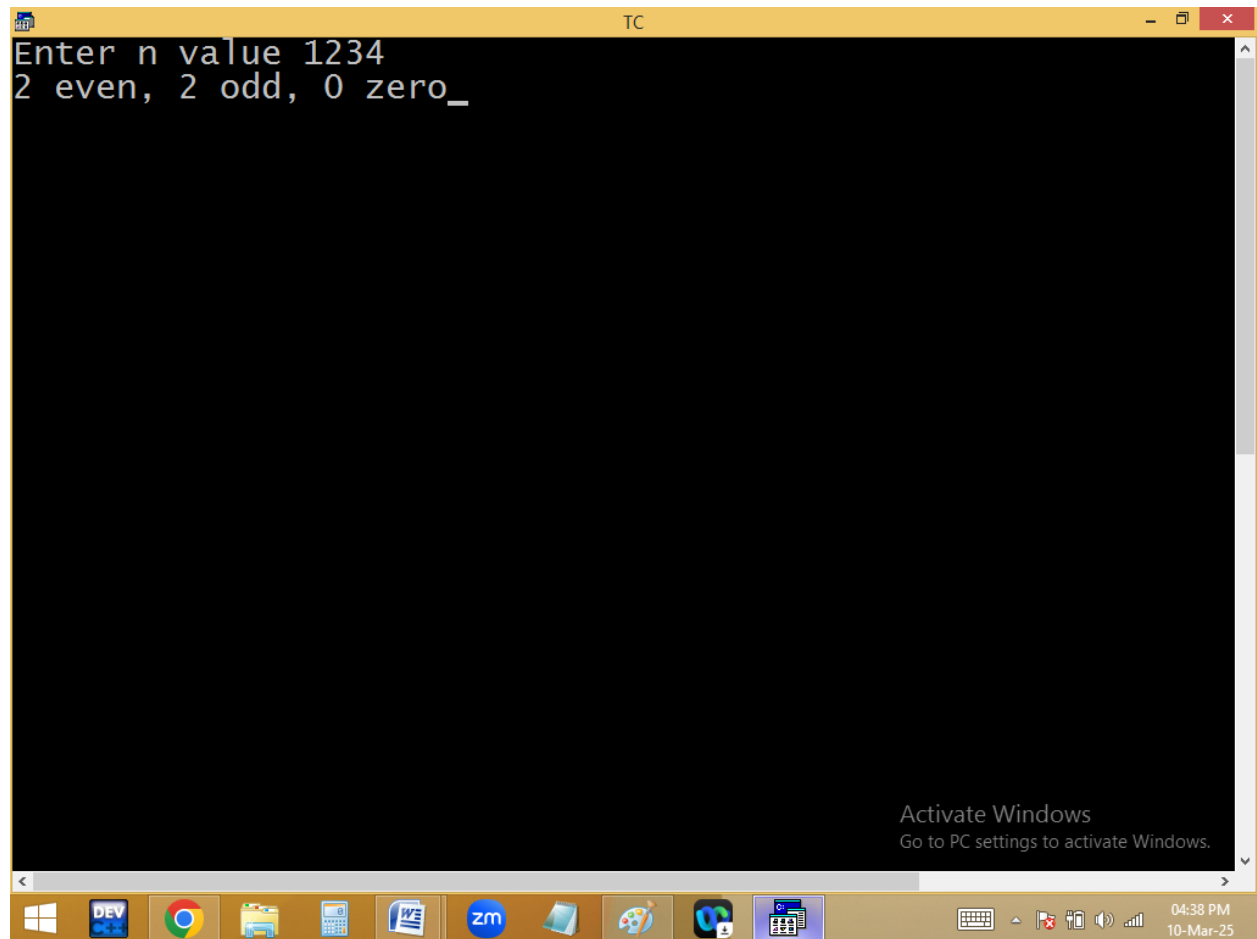
The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 14", "Col 9", and "Insert Indent Tab Fill Unindent *". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    long int n; int r, e,o,z;
    clrscr();
    printf("Enter n value "); scanf("%ld",&n);
    e=o=z=0;
    do
    {
        r=n%10;
        if(r==0)z++;else if(r%2==0)e++; else o++;
        n/=10;
    }while(n);
    printf("%d even, %d odd, %d zero",e,o,z);
    getch();
}
```

An "Activate Windows" watermark is visible in the bottom right corner of the IDE window, stating "Go to PC settings to activate Windows." Below the IDE window is the Windows taskbar, which includes icons for Windows, DEV, Chrome, File Explorer, Calculator, Word, Zoom, and other applications. The system tray on the right shows the time as "04:37 PM" and the date as "10-Mar-25". A secondary menu bar at the very bottom of the taskbar lists function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run".







```

Enter n value -1002
1 even, 1 odd, 2 zero

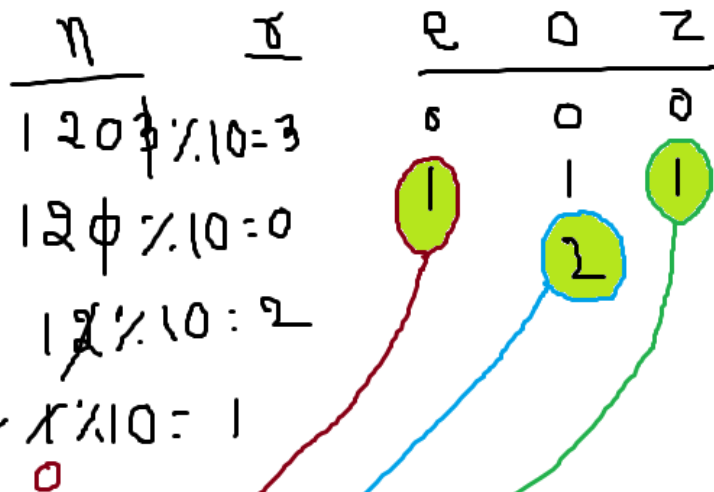
```

Activate Windows
Go to PC settings to activate Windows.

```

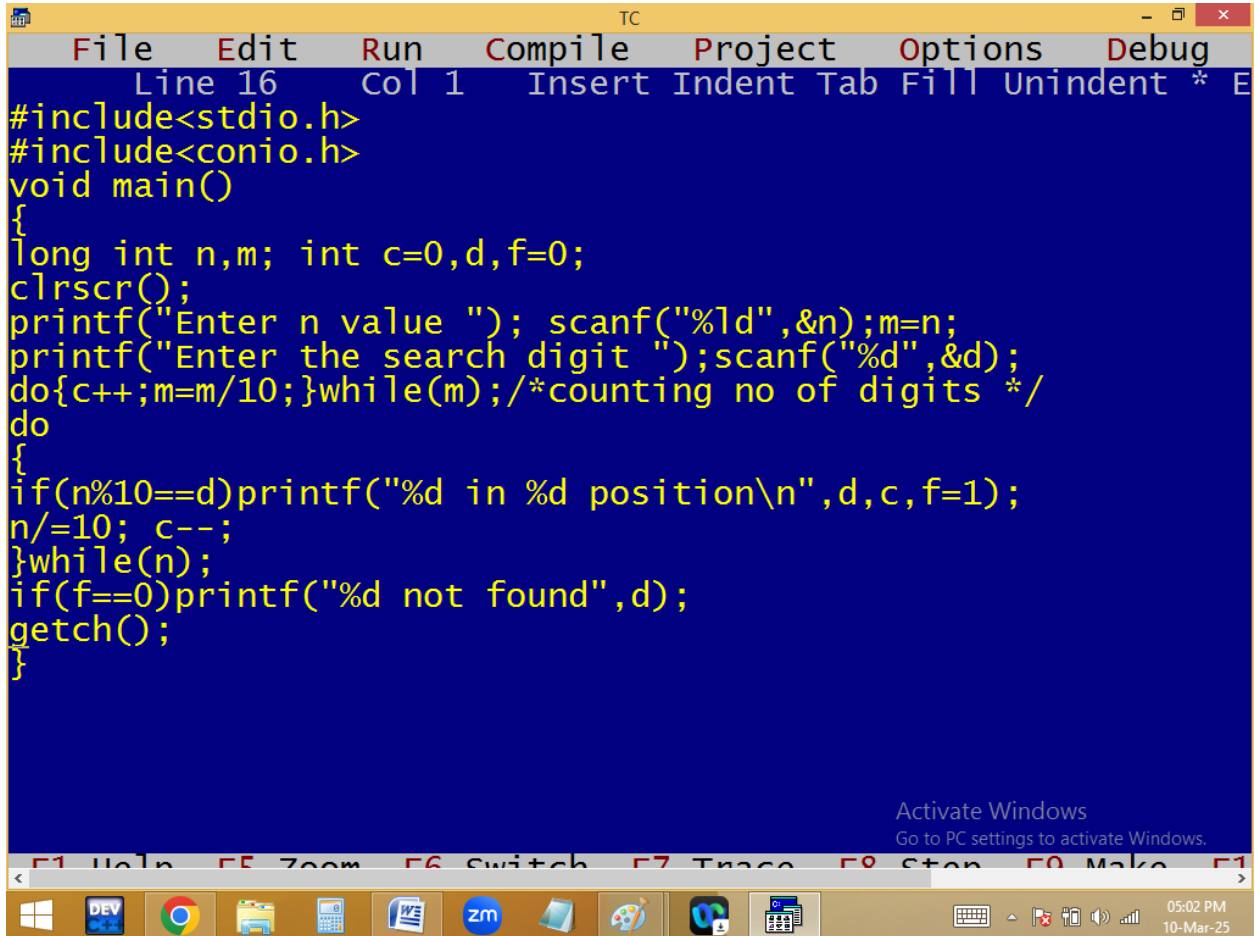
do
{
r=n%10;
if(r==0)z++;
else if(r%2==0)e++;
else o++;
n/=10;
}while(n);
p(e, o, z);

```



Finding the given digit position in given no

eg: 107347 → 7 position is 3, 6

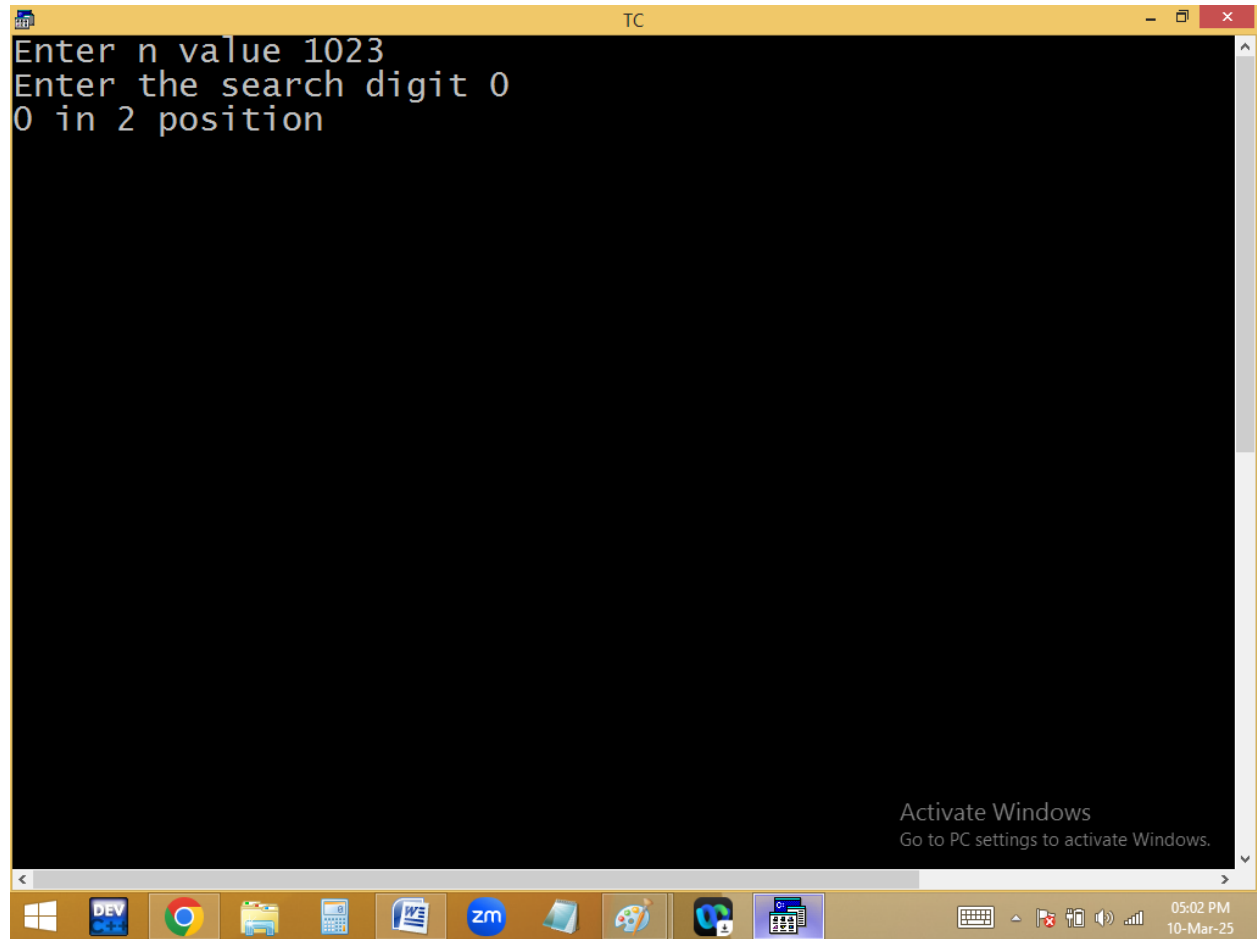


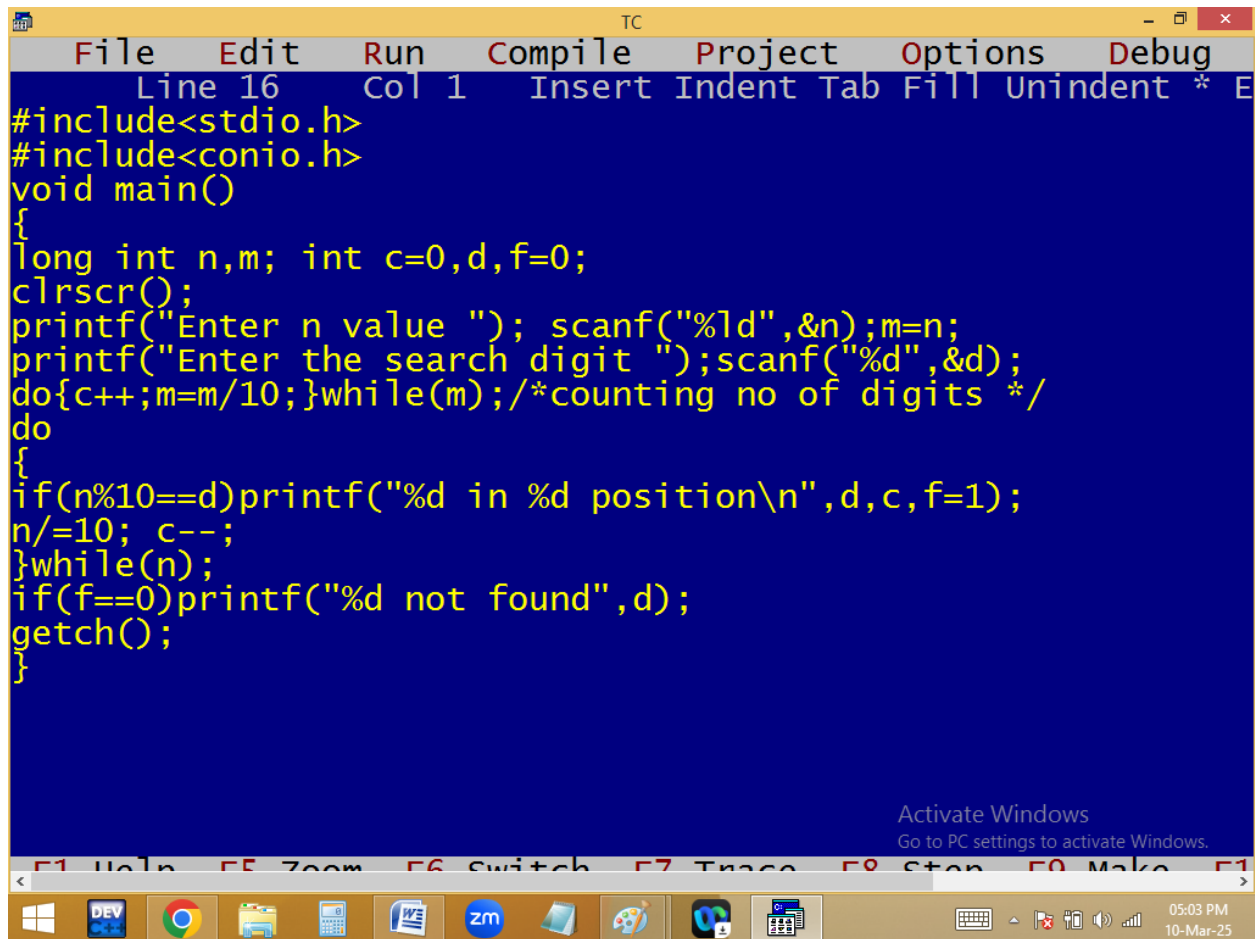
```
TC
File Edit Run Compile Project Options Debug
Line 16 Col 1 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
long int n,m; int c=0,d,f=0;
clrscr();
printf("Enter n value "); scanf("%ld",&n);m=n;
printf("Enter the search digit ");scanf("%d",&d);
do{c++;m=m/10;}while(m);/*counting no of digits */
do
{
if(n%10==d)printf("%d in %d position\n",d,c,f=1);
n/=10; c--;
}while(n);
if(f==0)printf("%d not found",d);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

05:02 PM
10-Mar-25

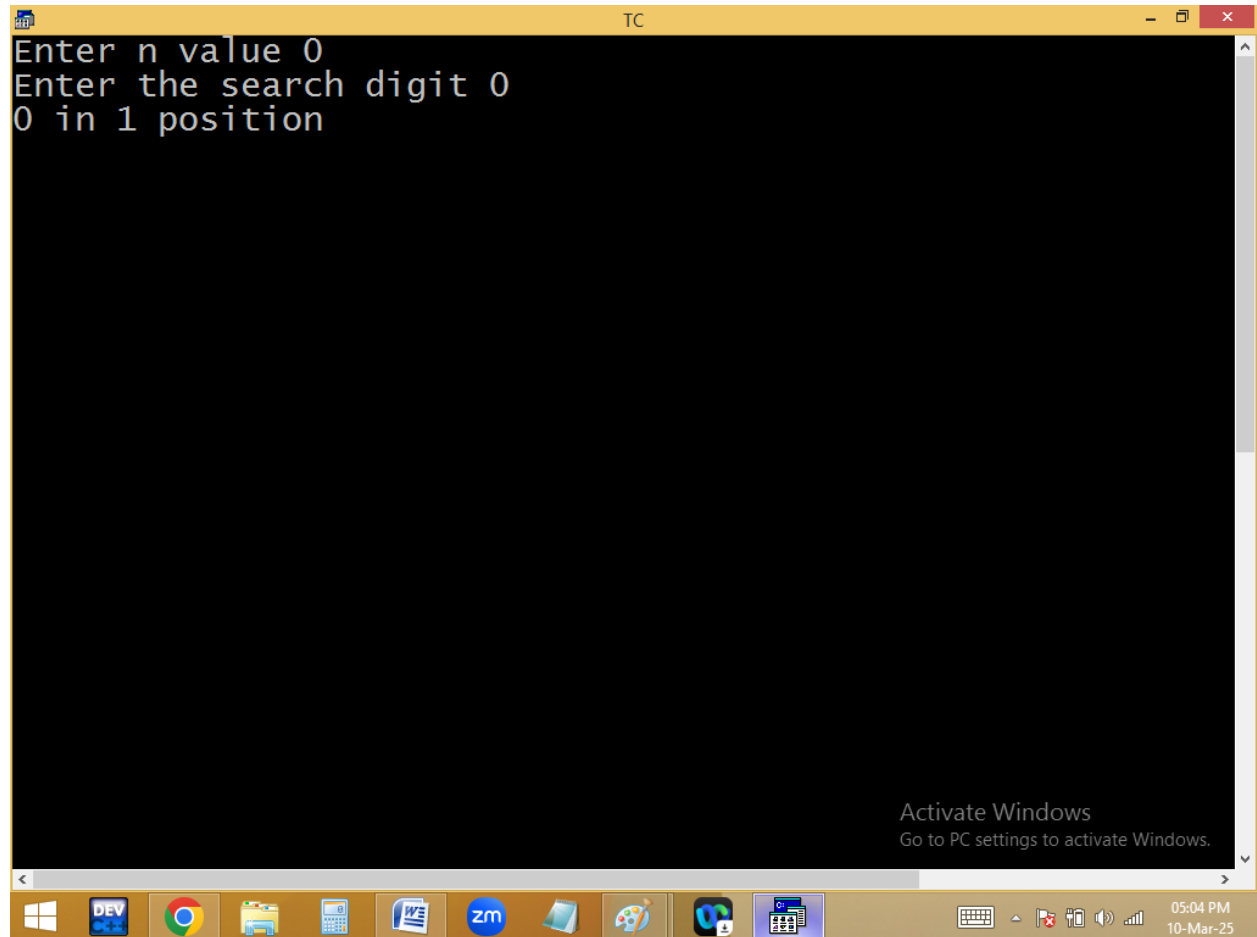


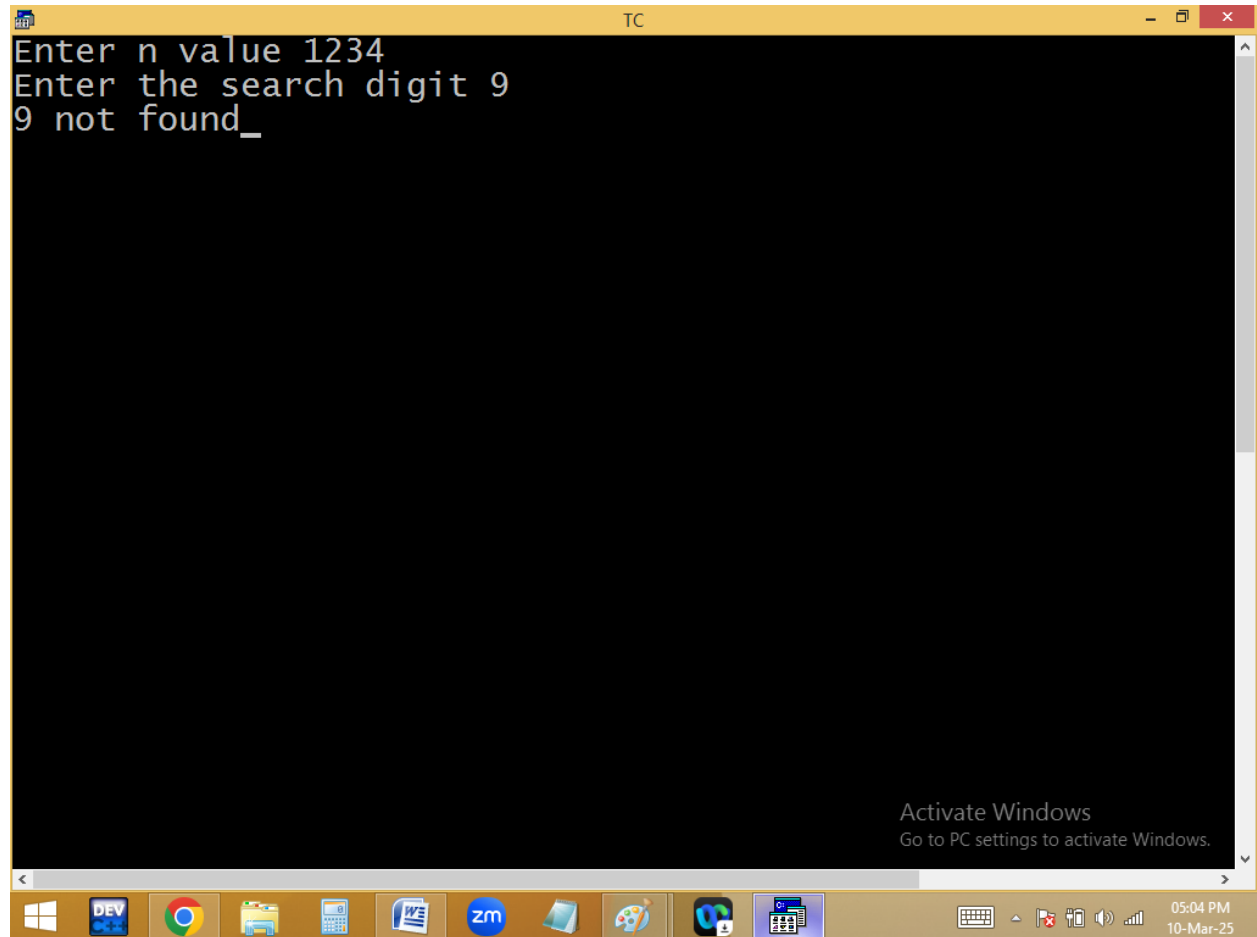


The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, the status bar shows "Line 16", "Col 1", and "Insert Indent Tab Fill Unindent * E". The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    long int n,m; int c=0,d,f=0;
    clrscr();
    printf("Enter n value "); scanf("%ld",&n);m=n;
    printf("Enter the search digit ");scanf("%d",&d);
    do{c++;m=m/10;}while(m);/*counting no of digits */
    do
    {
        if(n%10==d)printf("%d in %d position\n",d,c,f=1);
        n/=10; c--;
    }while(n);
    if(f==0)printf("%d not found",d);
    getch();
}
```

At the bottom of the IDE window, there is a toolbar with icons for various functions. Below the toolbar, a Windows taskbar is visible with icons for Windows, DEV, Chrome, File Explorer, Calculator, Word, Zoom, and other applications. The system tray on the right shows the date and time: "05:03 PM 10-Mar-25".





c = p(1057);

c	n	d	f
1	1057:7=150	0	0
2	1057%10=5	5	1
✓ 2	10%10=0	0	0

while(n)

{

if(n%10==d) p("%d in %d position\n",d,c,f=1);

n/=10;

c--;

}

if(f==0) p("%d not found", d);

do	m	c	n	d	f
✓ { c++; m=m/10; ✓ 0 1 2 3 4	0 1 2 3 4	0 1 2 3 4	1023%10=3	0	0
}while(m!=0);	3	3	102%10=2	0	1
do	2	2	10%10=0	0	0
{	1	1	1%10=1	0	1
	0	0		0	0

if(n%10==d) p("%d in %d position", d, c,f=1);

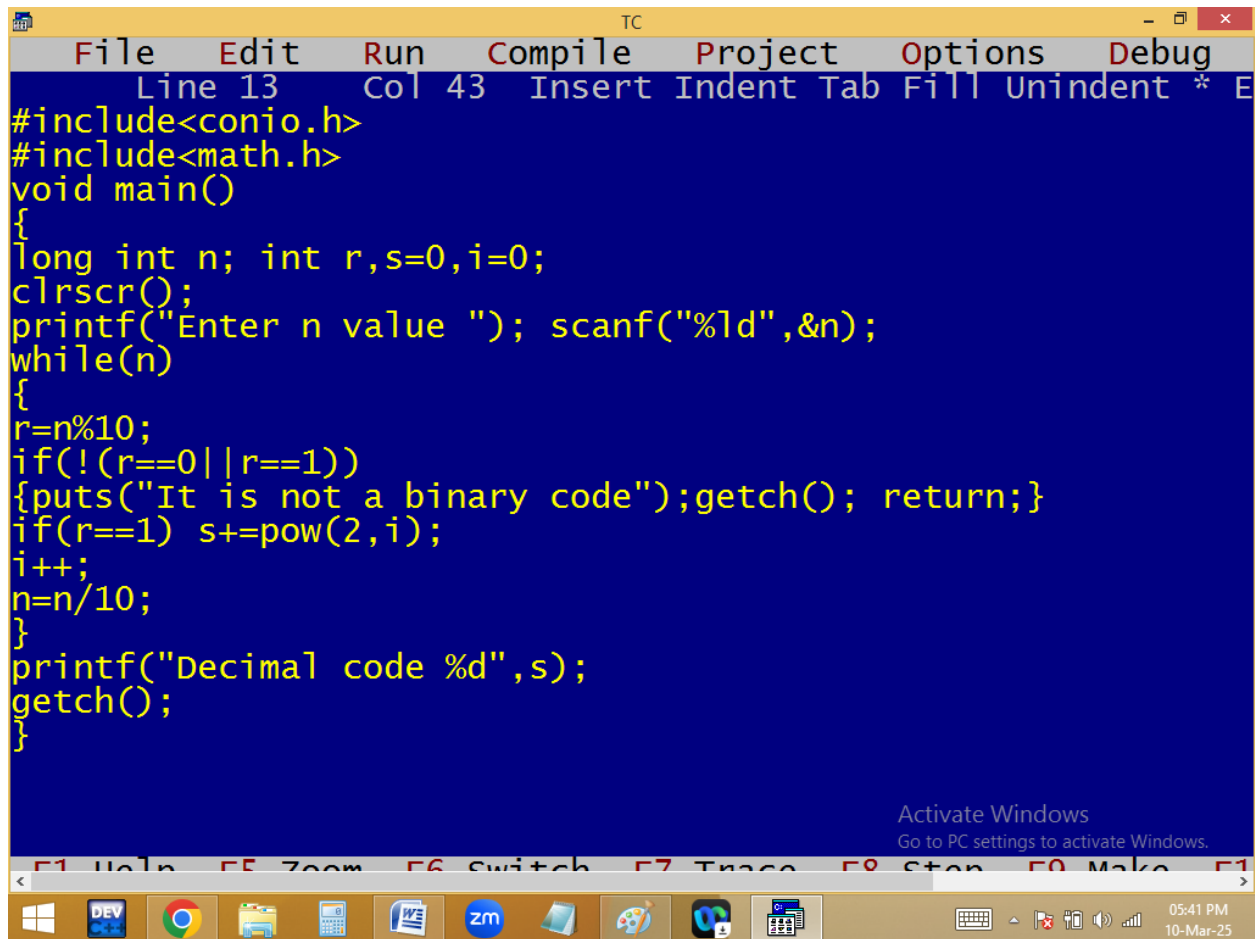
n=n/10; c--;

}while(n!=0);

if(f==0)p("%d not found",d);

Binary to Decimal conversion:

$$10100 = 20$$

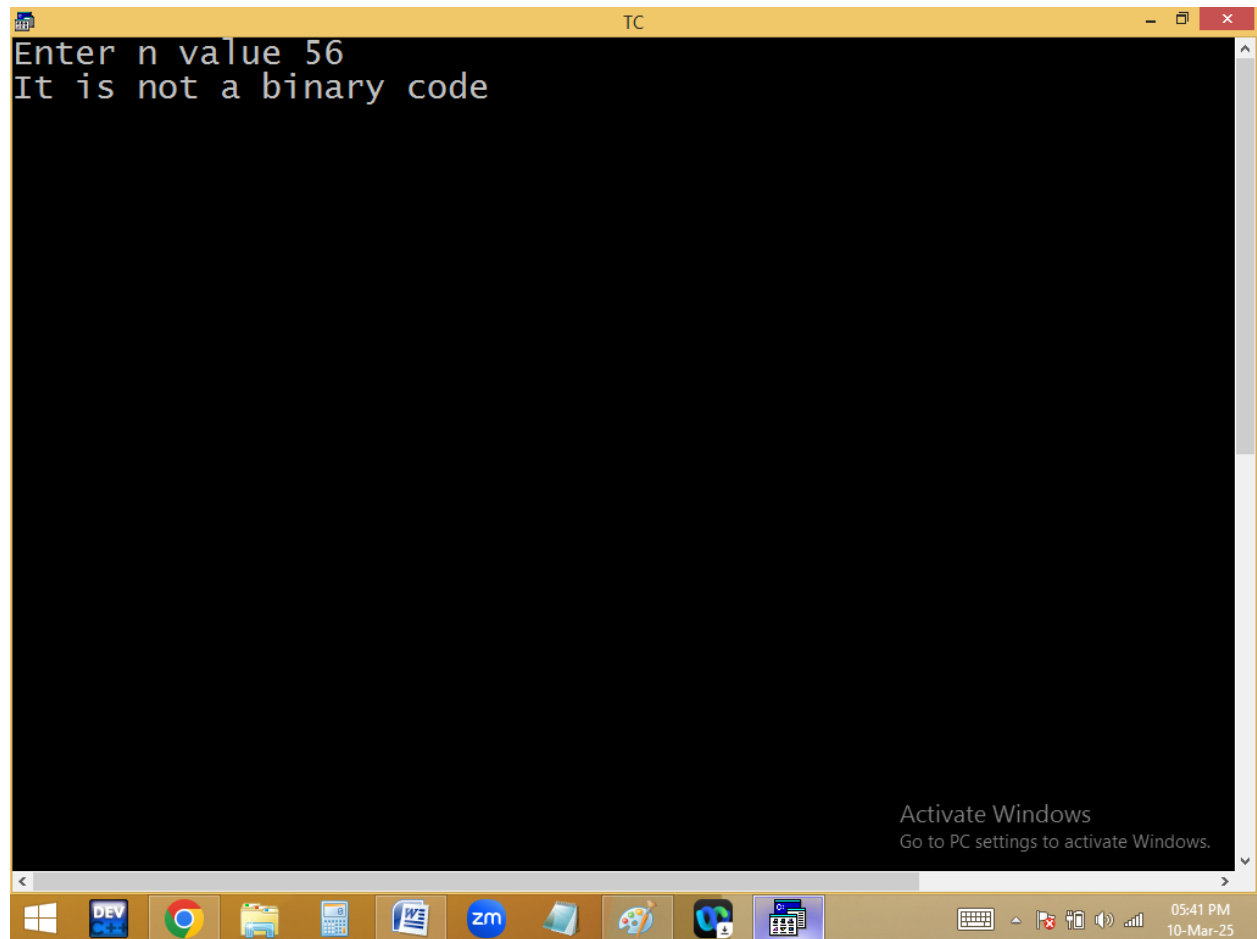


```
TC
File Edit Run Compile Project Options Debug
Line 13 Col 43 Insert Indent Tab Fill Unindent * E
#include<conio.h>
#include<math.h>
void main()
{
long int n; int r,s=0,i=0;
clrscr();
printf("Enter n value "); scanf("%ld",&n);
while(n)
{
r=n%10;
if(!(r==0||r==1))
{puts("It is not a binary code");getch(); return;}
if(r==1) s+=pow(2,i);
i++;
n=n/10;
}
printf("Decimal code %d",s);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

Windows taskbar: DEV, Chrome, File Explorer, Calculator, Word, Zoom, Paint, VS Code, Task Manager, System Tray (Keyboard, Volume, Network, Date/Time: 05:41 PM 10-Mar-25)



```
Enter n value 10100
Decimal code 20
```

Activate Windows
Go to PC settings to activate Windows.

<pre>i=0; while(n!=0) { r=n%10; if(r!=0 r!=1) if(r==1)s+=pow(2,i); i++; n=n/10; } p(s);</pre>	$\begin{array}{r} i \\ \hline 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{array}$	$\begin{array}{r} n \\ \hline 10100 \div 10 = 0 \\ \begin{smallmatrix} 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \end{smallmatrix} \\ 1010 \div 10 = 0 \\ 101 \div 10 = 1 \\ 10 \div 10 = 0 \\ 1 \div 10 = 1 \\ 0 \end{array}$	$\begin{array}{r} s \\ \hline 0 \\ + 4 \\ + 16 \\ \hline 10 \end{array}$
--	---	--	--